



INITIAL QUBIT MAPPING IN PAULI POLYNOMIAL SYNTHESIS

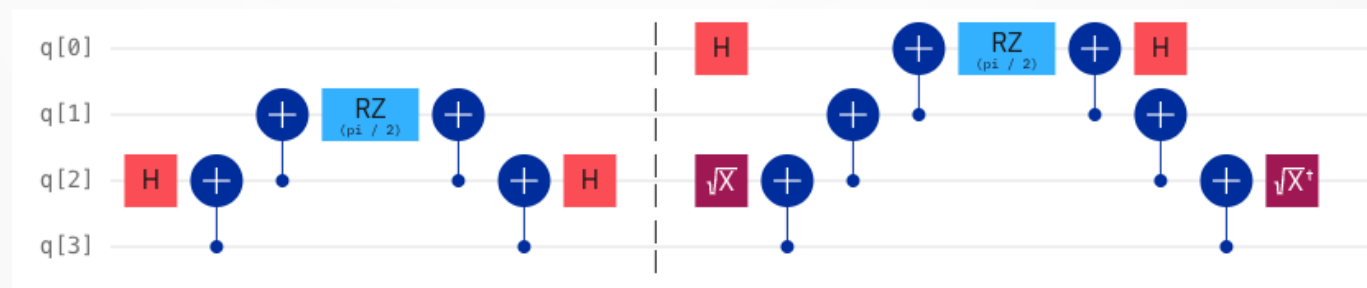
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INTRODUCTION TO THE TOPIC

- Pauli polynomials (composed of several pauli strings) are widely used for example in VQE algorithms.
- Naïve synthesis to quantum circuits can be done trivially but it produces lot of CNOTs
- How to synthesize pauli polynomials so that cnot count is minimized?

$\alpha\beta$
 q_0 IX
 q_1 ZZ
 q_2 XY
 q_3 ZZ





PREVIOUS WORK

- Here we focus to synthesis of trotterized Hamiltonians time-evolution operators
 - Order of pauli strings can be changed with cost of only small error
- Algorithm should take qubit connectivity into account
- This type of algorithm was described in paper 2024: “Redefining Lexicographical Ordering: Optimizing Pauli String Decompositions for Quantum Compiling”
 - Algorithm is significantly better than Paulihedral and TKET on randomized circuits and different molecular ansatzes.
 - Synthesis produces to circuit having two parts: leading part having H,S,V,RZ and CNOT gates, and trailing part having H, $S^\dagger$, $V^\dagger$ and CNOT gates. Trailing part can be condensed with Clifford tableau
 - Algorithm uses trivial qubit mapping, could there be better solution?



NEW SYNTHESIS ALGORITHM (DYNAMIC ORDERING)

- Order of processing pauli strings:
 - Pauli strings are processed in order of size: pauli string having least non-I paulis is processed next. Secondary determining factor is total length of pauli chain.
 - When pauli string is processed we must propagate trailing gates through all remaining pauli strings.
 - Propagation can alter size of pauli strings, which influences on which paulistring is to be processed next.

	2	2	0	2	1	3	1	0	2	3	3	1	1	1	3	3	0	0	0	1	0	0	1	2	3	0	1	2	3	0	2	1	1	2	2	0	2
	4	8	3	7	7	3	1	4	0	2	6	4	2	9	0	4	0	8	5	0	1	6	6	9	1	9	5	1	5	2	6	8	3	3	5	7	2
0			Y									X	X						Z		Z	Z	X		Z	X	Y			Z	X	Y	X	Y	Z	X	
1		X							X	Z			Z	Z			Y		X	X	Z	Y	Z	Z		Z			Y	Y	X			X			
2											X	Z	X	Y	X	Z		X		Y	Y	Y	Z	Y	X		Z					Z			Y		
3				Z		X	Z	Z				X	Z	Z	Z	X	Z		X	Z	Z			Z	Y		X		X			Z					
4	X			Z	Z	X			Z		Y		Y	X	Z		X	X	Y		X	X	X	X	Y	Y	X			Z	X			Z			
5			Z		Y	X							X	Y	Z		Y		X				Y				X	X		X	X	Z	X	Z	Y		



NEW SYNTHESIS ALGORITHM (DYNAMIC ORDERING)

- Order of processing paulis in pauli string:
 - For example line topology has two possibilities to remove next pauli: two ends of chain
 - Removing pauli can be done with several gate combinations: I, H, S, V, HS or SH in both qubit and CNOT having qubit1 or qubit2 as a target qubit. So together 72 different options.
 - We choose end and gate combination which has minimum net effect to the length of remaining paulistringss. Secondary criteria is ability to remove I from the center of chain (these take two cnots to process)

	2	1	2	1	1	2	0	1	0	1	1	2	0	0	1	0	0	2	1	1	1	
	4	7	0	1	5	7	2	3	6	0	6	6	1	5	2	0	9	2	8	9	4	
0					X	Y		Z	Y	Y	Y	Z		X	X	X	Y	Z	X			
1					X	X	X							X	Y		Y				X	
2										Y	X	Y		Z	Y	Z	Z	Y	Y	X	Y	X
3								Z	Z	Y	X	Y		X	Z	Y	Z	Z	X		X	
4										X	Y	X		Y	Y	Y	Z		X	Z	Z	Z
5										Z	Y	Y	Y	Y	Y	Y	X	Z		Z	X	X

	2	1	2	1	1	2	0	1	0	1	1	2	0	0	1	0	0	2	1	1	1
	4	7	0	1	5	7	2	3	6	0	6	6	1	5	2	0	9	2	8	9	4
0						X	Y		Z	Y	Y	Y	Z		X	X	X	Y	Z	X	
1						X	X	X							X	Y		Y			X
2																					
3																					
4																					
5																					



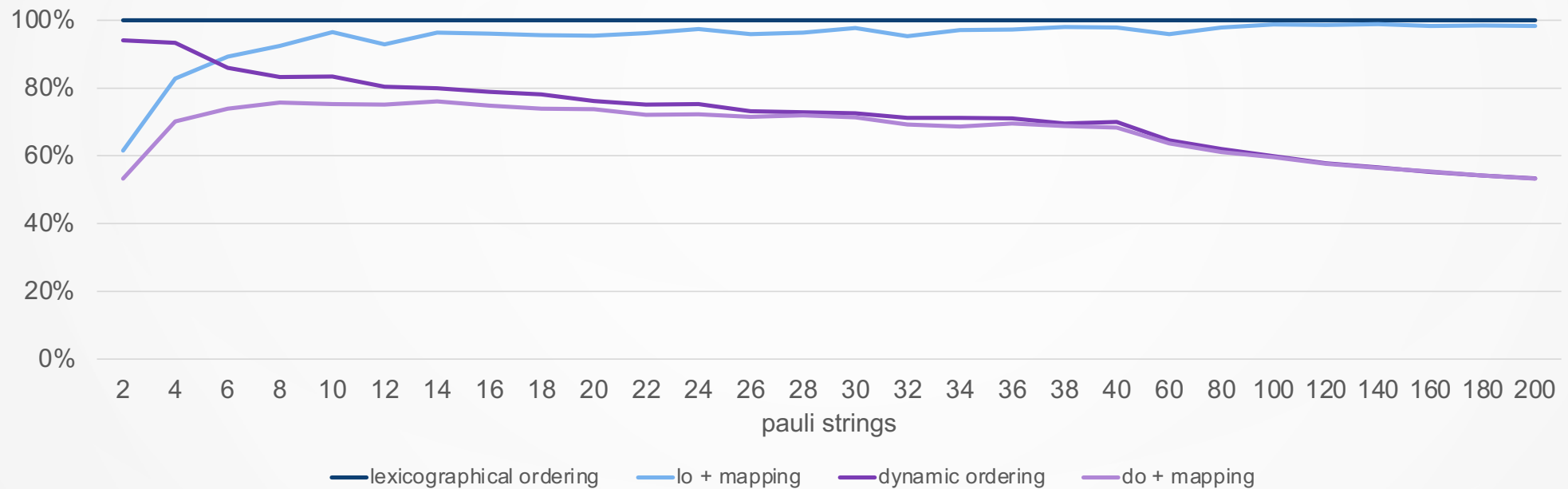
MAPPING ALGORITHM

- Qubit-pairs are scored according to number of pauli strings which have non-I paulis in both qubits
- Central physical qubit is identified: qubit which have longest distance to the nearest qubit having only one connection.
- Heaviest logical qubit (qubit having most non-I paulis) is mapped to central physical qubit.
- Non-mapped physical qubit is chosen from the neighbors of mapped physical qubit. Logical qubit having max score with mapped logical qubit is selected
- This algorithm decreases the number of I paulis in the middle of chain, leads to kind of “compact chains”



RESULTS

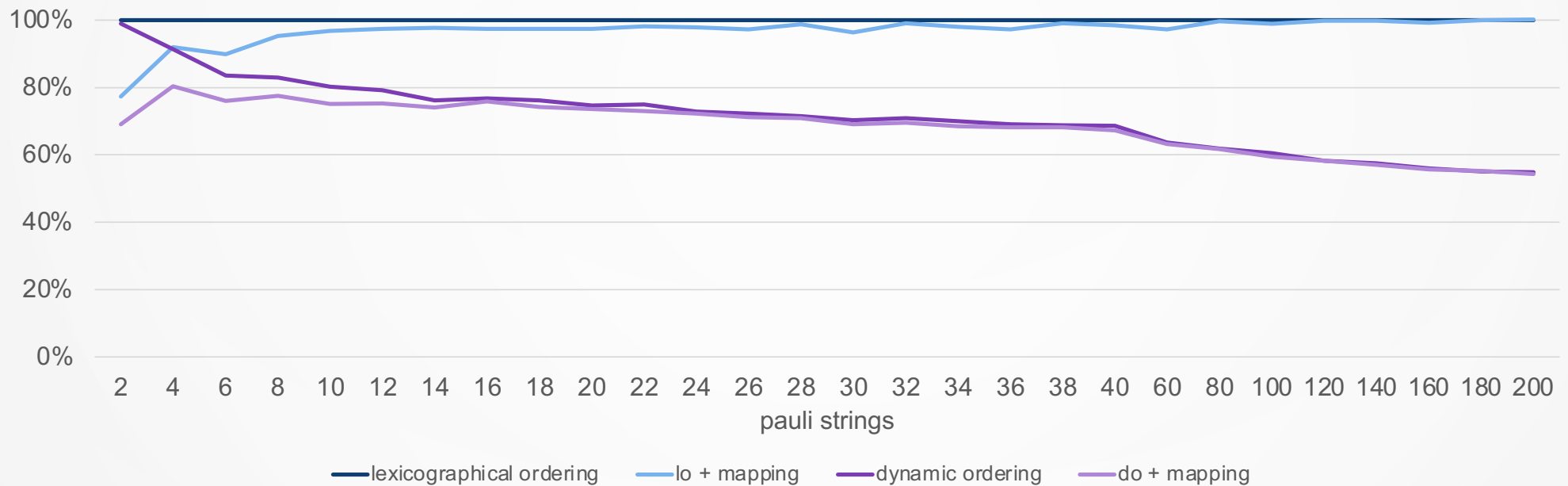
Line-6 (cnot count relative to lexicographical ordering without mapping, 200 samples)





RESULTS

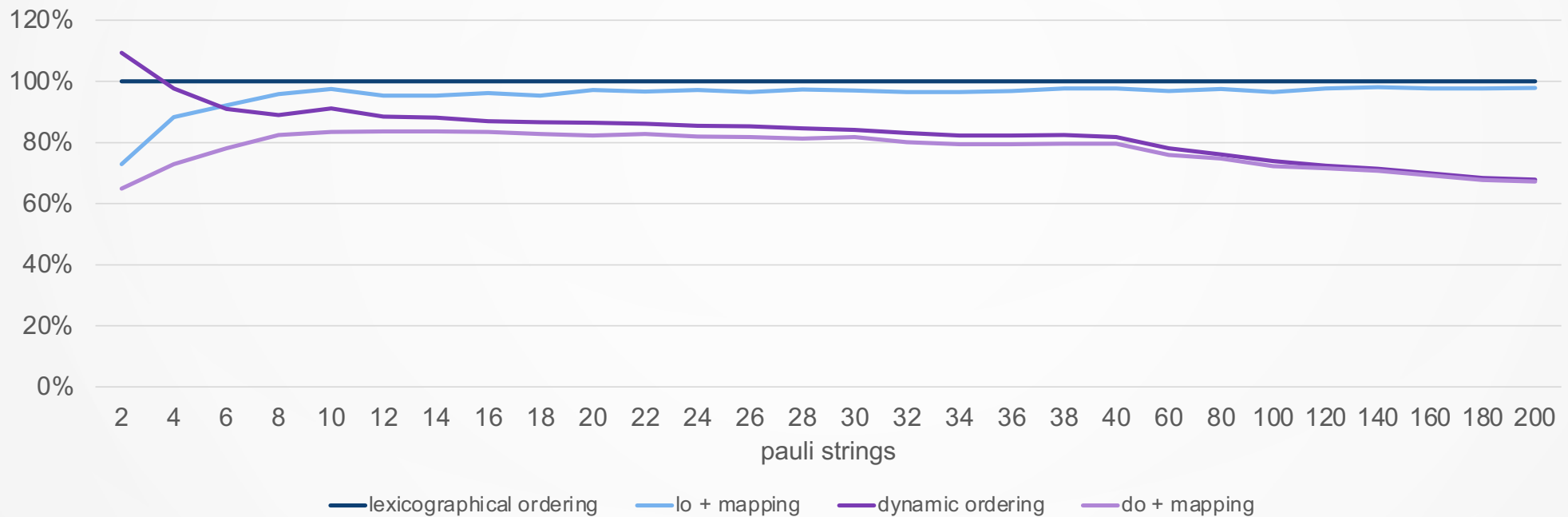
Quito-5 (cnot count relative to lexicographical ordering without mapping, 200 samples)





RESULTS

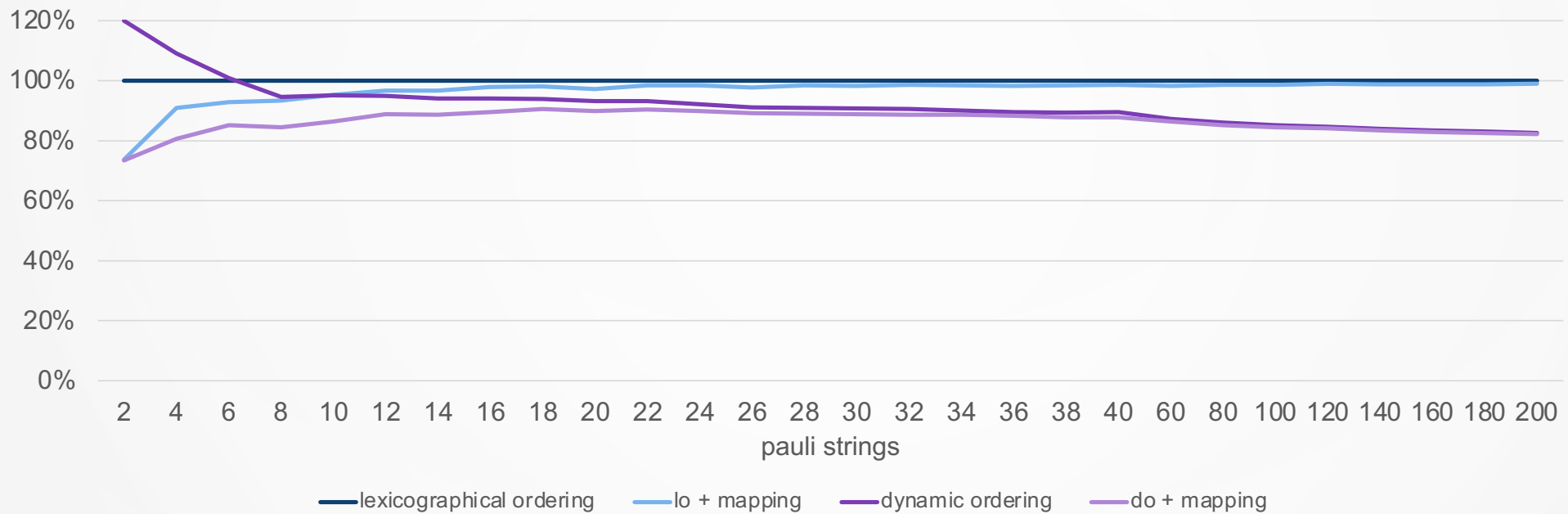
Grid-9 (cnot count relative to lexicographical ordering without mapping, 200 samples)





RESULTS

Guadalupe-16 (cnot count relative to lexicographical ordering without mapping, 200 samples)





RESULTS

molecule	backend	qubits	pauli s.	cx lo	lo>do	do->dom
BeH2_BK_sto3g	guadalupe	14	1488	12419	-36%	36%
BeH2_JW_sto3g	guadalupe	14	1488	12181	-58%	60%
BeH2_P_sto3g	guadalupe	12	1085	8482	-57%	-17%
C2_P_sto3g	mumbai	18	2950	37786	-40%	-34%
CH2_BK_sto3g	guadalupe	14	1488	12919	-46%	31%
CH2_JW_sto3g	guadalupe	14	1488	13433	-20%	-38%
CH2_P_sto3g	guadalupe	12	2109	14909	-67%	-3%
H2O_BK_sto3g	guadalupe	14	1000	8675	-48%	57%
H2O_JW_sto3g	guadalupe	14	1000	7967	-37%	-31%
H2O_P_sto3g	guadalupe	12	1085	8077	-61%	-8%
H2_BK_631g	guadalupe	8	84	443	-33%	-9%
H2_BK_sto3g	quito	4	12	35	-37%	9%
H2_JW_631g	guadalupe	8	84	414	-23%	-16%
H2_JW_sto3g	quito	4	12	38	-37%	21%
H2_P_631g	nairobi	6	158	433	-46%	-9%
H2_P_sto3g	quito	2	4	5	0%	-60%

molecule	backend	qubits	pauli s.	cx lo	sg->do	do->dom
H4_BK_631g	mumbai	16	1440	13852	-16%	-3%
H4_BK_sto3g	guadalupe	8	160	878	-49%	-16%
H4_JW_631g	mumbai	16	1440	11727	0%	-42%
H4_JW_sto3g	guadalupe	8	160	883	-41%	-21%
H4_P_631g	guadalupe	14	2912	24224	-34%	-49%
H4_P_sto3g	nairobi	6	164	471	-54%	-12%
H8_BK_sto3g	mumbai	16	2688	28018	-26%	-43%
H8_JW_sto3g	mumbai	16	2688	29280	-69%	17%
HCl_BK_sto3g	mumbai	20	684	6308	-24%	21%
HCl_JW_sto3g	mumbai	20	684	5807	-16%	5%
LiH_BK_631g	mumbai	22	3240	35792	11%	16%
LiH_BK_sto3g	guadalupe	12	640	4250	-39%	20%
LiH_JW_631g	mumbai	22	3240	33673	-20%	-41%
LiH_JW_sto3g	guadalupe	12	640	4331	-48%	-8%
LiH_P_sto3g	guadalupe	10	630	3472	-58%	-4%
N2_P_sto3g	mumbai	18	2950	35602	-53%	-9%
NH3_BK_sto3g	mumbai	16	2340	25280	-22%	-42%
NH3_JW_sto3g	mumbai	16	2340	23182	-63%	9%
NH_BK_sto3g	guadalupe	12	640	4788	-54%	-8%
NH_JW_sto3g	guadalupe	12	640	4884	-50%	-4%
NH_P_sto3g	guadalupe	10	630	3472	-58%	-4%



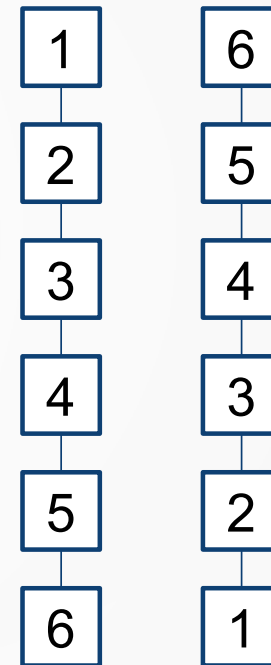
DISCUSSION

- Results:
 - Mapping algorithm is good when number of pauli strings is low. Increasing randomness makes qubit mapping less influential
 - Dynamic ordering is better than lexicographical ordering, difference increases when number of pauli strings is increased
- Randomness:
 - Synthesis algorithm has significant built in randomness: both pauli string ordering and choosing order of paulis to remove have equal options which algorithms just chooses one
 - This randomness has almost as much contribution to cnot count as mapping.
 - Optimal mapping in one set of choices is poor mapping in other set of choices
 - Mapping must be adjusted according how synthesis algorithm functions?



DISCUSSION

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 - This randomness has almost as much contribution to cnot count as mapping.
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DISCUSSION

- Synthesis algorithm resembles algorithm published in 2024:
<https://arxiv.org/abs/2404.03280>
 - Similar idea for scoring gate-combination possibilities, how ever more limited
 - two major differences: 1) does not consider topology, 2) does not allow ordering
- Mapping algorithm resembles algorithm published in 2025:
<https://ieeexplore.ieee.org/abstract/document/10771974>
 - Almost the same. Small difference, which does not have real impact